

Deep Evidential Regression for Flood-Impact Prediction in Urban Transport

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Abstract. Climate-adaptation planning increasingly relies on simulation-based models to evaluate intervention decisions. Using MA’AT flood-transport trajectories, we learn neural surrogates to predict remaining cumulative transport cost while studying their reliability under intervention-policy shift. We compare deterministic regression, heteroscedastic Gaussian regression, MC Dropout, deep ensembles, and Deep Evidential Regression (DER) over 20 training seeds, with checkpoints selected by validation RMSE. MC Dropout achieves the strongest predictive accuracy, with RMSE $(2.191 \pm .337) \times 10^6$, while its predictive standard deviation achieves AUROC $.9671 \pm .0058$ for detecting the random-to-no-intervention shift. Although DER gives weaker point predictions, its evidential uncertainty signals respond differently to the same shift: the log-transformed NIG variance-scale parameter ($\log \beta$) achieves AUROC $.9666 \pm .0344$, while other NIG-derived uncertainty signals show markedly different shift-detection performance. These results show that predictive accuracy and shift sensitivity are not necessarily aligned, and that DER reliability depends strongly on the chosen evidential uncertainty signal.

1 Introduction

Urban flooding can disrupt transport networks and increase mobility costs. Climate-adaptation planning therefore requires tools for evaluating interventions before deployment. We study this problem using MA’AT, which integrates flood modelling, urban transport simulation, impact estimation, and reinforcement learning [3, 4]. From MA’AT trajectories, we learn a surrogate that predicts the remaining cumulative transport cost from the current state and intervention action. A surrogate trained under random interventions may, however, become unreliable under a different policy such as no intervention. Point predictions alone cannot reveal this shift, motivating uncertainty-aware surrogate modelling. Deep evidential regression (DER) [1] predicts a Normal-Inverse-Gamma (NIG) distribution in a single forward pass. However, prior work has shown that evidential regression can be sensitive to overparameterization, optimization dynamics, and the parameterization and interpretation of evidence [2, 7, 8]. Moreover, DER outputs several NIG parameters from which different uncertainty signals can be derived, and it is unclear whether these signals respond similarly to policy shift. Here, an evidential signal refers to an uncertainty score computed from

these predicted NIG parameters, such as aleatoric uncertainty, epistemic uncertainty, predictive scale, or individual parameter-based scores. We therefore first examine multiple uncertainty signals derived from the same trained DER model. We then compare DER with deterministic regression, Gaussian regression, MC Dropout [5], and deep ensembles [6] under the same intervention-policy shift. Our results show that predictive accuracy and shift sensitivity are not necessarily aligned. For DER, the choice of evidential signal is particularly important. Our contributions are: (1) formulating intervention-policy shift as an uncertainty problem for MA'AT surrogate prediction; (2) comparing deterministic, probabilistic, stochastic, ensemble, and evidential surrogates under a common multi-run protocol; and (3) showing that shift-detection performance can vary substantially across uncertainty signals derived from the same DER model.

2 Deep Evidential Learning

Let $\mathcal{D}_{\text{tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the simulated samples used for training. Each sample corresponds to one time step of a MA'AT episode, where $\mathbf{x}_i$ contains the current observation and intervention action, and y_i is the cumulative transport cost from that step until the end of the episode. In the context of this study, training and in-distribution (ID) trajectories follow a random intervention policy, while distribution-shift evaluation uses a deterministic no-intervention policy.

Deep Evidential Regression. We use DER to model uncertainty in the flood-impact transport surrogate. For each input $\mathbf{x}$, the network predicts the parameters of a NIG distribution,

$$(\gamma, \nu, \alpha, \beta) = f_{\theta}(\mathbf{x}), \quad (1)$$

where γ represents the predictive mean, while ν , α , and β characterize the predictive uncertainty. The parameters satisfy $\nu > 0$, $\alpha > 1$, and $\beta > 0$, enforced using softplus transformations. The point prediction is $\hat{y} = \gamma$. The model is trained using the standard DER objective,

$$\mathcal{L}_{\text{DER}} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{evi}} \mathcal{L}_{\text{reg}}, \quad (2)$$

where $\mathcal{L}_{\text{NLL}}$ is the NIG negative log-likelihood and $\mathcal{L}_{\text{reg}}$ penalizes high evidence for inaccurate predictions. We use $\lambda_{\text{evi}} = 10^{-3}$.

Evidential uncertainty signals. Rather than assuming that a single NIG quantity is universally suitable for distribution-shift detection, we examine several uncertainty signals derived from the same trained DER model:

$$\begin{aligned} u_{\text{alea}} &= \frac{\beta}{\alpha - 1}, & u_{\text{epis}} &= \frac{\beta}{\nu(\alpha - 1)}, & u_{\text{pred}} &= \sqrt{\frac{\beta(1 + \nu)}{\nu(\alpha - 1)}}, \\ u_{\text{St}} &= \sqrt{\frac{\beta(1 + \nu)}{\alpha\nu}}, & u_{\nu} &= \frac{1}{\sqrt{\nu}}, & u_{\beta} &= \log \beta. \end{aligned} \quad (3)$$

Here, u_{alea} and u_{epis} are the standard NIG aleatoric and epistemic variances, u_{pred} is the predictive standard deviation, u_{ν} is the inverse mean-evidence signal,

u_{st} is the Student- t predictive scale, and u_{β} is the log-transformed NIG variance-scale parameter. All signals are computed post hoc from the same trained DER checkpoint, so detection differences reflect only the extracted uncertainty quantity, not model architecture or training.

3 Experiments

Data and protocol. We use samples generated from MA’AT flood-transport simulation episodes, with observations and intervention actions as inputs and remaining cumulative episode cost as the target. The dataset contains 20 random-policy and 20 duplicate no-intervention episodes, each with 77 time-step samples. We retain one unique no-intervention trajectory for policy-shift evaluation and split the random-policy episodes into 14/3/3 train/validation/test episodes. Inputs have 1,664 dimensions. The episode split is fixed across all experiments to avoid trajectory overlap between training and evaluation. All methods are evaluated over 20 training seeds, and results are reported as mean $\pm$ standard deviation. For each run, the checkpoint with minimum validation RMSE is selected without using the shifted trajectory.

Baselines and metrics. We compare a deterministic MLP, heteroscedastic Gaussian regression, MC Dropout, a deep ensemble, and Deep Evidential Regression (DER). Prediction quality is evaluated using MAE, RMSE, and R^2 on held-out random-policy episodes; policy-shift detection uses AUROC, AUPR, and FPR95, with random-policy test samples as ID and the no-intervention trajectory as shifted. The non-evidential baselines use a 256–128–64 ReLU MLP with batch normalization after the first layer and dropout rates .20/.10, and are trained with Adam using learning rate 10^{-3} , batch size 64, and up to 200 epochs. DER is allowed up to 300 epochs to accommodate potentially slower optimization of the evidential objective; all selected DER checkpoints occurred before epoch 45. The Gaussian model uses heteroscedastic Gaussian NLL. MC Dropout uses predictive standard deviation over 50 stochastic passes, while the ensemble uses disagreement across 5 independently initialized members. For DER, we compare several uncertainty quantities derived from the same NIG prediction.

Table 1: Prediction accuracy on held-out random-policy episodes and detection of the no-intervention policy shift. Checkpoints use minimum validation RMSE; uncertainty is predictive std for MC Dropout, ensemble disagreement, and strongest DER signal.

Model	Score	MAE $\downarrow$	RMSE $\downarrow$	$R^2 \uparrow$	AUROC $\uparrow$	AUPR $\uparrow$	FPR95 $\downarrow$
MLP	—	2.058 $\pm$.311	3.062 $\pm$.569	.9936 $\pm$.0024	—	—	—
Gaussian	Pred.std	2.448 $\pm$.444	3.503 $\pm$.729	.9915 $\pm$.0035	.9571 $\pm$.0356	.7630 $\pm$.1385	.0437 $\pm$.0368
MC Dropout	Pred.std	1.544 $\pm$.189	2.191 $\pm$.337	.9968 $\pm$.0010	.9671 $\pm$.0058	.8117 $\pm$.0222	.069 $\pm$.0156
Deep Ensemble	Disagr.	1.748 $\pm$.146	2.485 $\pm$.257	.9959 $\pm$.0008	.7919 $\pm$.1112	.4488 $\pm$.1497	.2130 $\pm$.1130
DER	log β	3.742 $\pm$.973	5.592 $\pm$ 1.246	.9783 $\pm$.0103	.9666 $\pm$.0344	.8055 $\pm$.1311	.0342 $\pm$.0364

Results and discussion. Table 1 shows a clear distinction between predictive accuracy and policy-shift detection. MC Dropout provides the strongest point prediction among the uncertainty-aware baselines, achieving an RMSE of $(2.191 \pm .337) \times 10^6$ and $R^2 = .9968 \pm .001$. Its predictive standard deviation

Table 2: Comparison of uncertainty signals from the same DER model for detecting no-intervention policy shift. Results use the checkpoint selected by minimum validation RMSE. Differences reflect only the choice of evidential uncertainty signal.

Metric	$\frac{\beta}{\alpha-1}$	$\frac{\beta}{\nu(\alpha-1)}$	$1/\sqrt{\nu}$	t -scale	$\log \beta$
AUROC $\uparrow$	.9189 $\pm$.0750	.7858 $\pm$.0961	.1255 $\pm$.2624	.9210 $\pm$.0573	.9666$\pm$.0344
AUPR $\uparrow$	.6627 $\pm$.1524	.4328 $\pm$.1241	.1783 $\pm$.0998	.6528 $\pm$.1350	.8055$\pm$.1311
FPR95 $\downarrow$	.0833 $\pm$.0771	.2180 $\pm$.0979	.8794 $\pm$.2642	.0818 $\pm$.0590	.0342$\pm$.0364

also provides strong shift detection, with AUROC .9671 $\pm$.0058. The Gaussian model obtains slightly weaker predictive accuracy, but its predictive standard deviation remains highly sensitive to the shift, reaching AUROC .9571 $\pm$.0356. The deep ensemble achieves competitive prediction accuracy but substantially weaker shift detection, with AUROC .7919 $\pm$.1112. DER produces the weakest point prediction among the evaluated neural surrogates, with RMSE $(5.592 \pm 1.246) \times 10^6$. However, its strongest evaluated evidential signal, $\log \beta$, achieves AUROC .9666 $\pm$.0344, comparable to MC Dropout despite requiring only a single evidential forward pass at inference. To assess whether the policy shift is directly detectable from the intervention action, we additionally evaluate an action-only baseline using $\mathbb{I}[a = 0]$ as the shift score. It achieves AUROC .6883, AUPR .3484, and FPR95 .6234, substantially below Gaussian regression, MC Dropout, and DER $\log \beta$. Thus, although the deterministic no-intervention policy induces a detectable shift in the action component, the strong uncertainty-based separation cannot be explained by the action feature alone. Table 2 further shows that policy-shift sensitivity depends strongly on which NIG quantity is used as uncertainty. The ν -based signal $1/\sqrt{\nu}$ performs poorly, with AUROC .1255 $\pm$.2624, while quantities involving β provide substantially stronger separation. Aleatoric uncertainty and the Student- t predictive scale reach AUROCs of .9189 $\pm$.0750 and .9210 $\pm$.0573, respectively, while $\log \beta$ yields the strongest observed separation. These results indicate that evidential parameters can respond very differently to intervention-policy shift and that uncertainty quality cannot be inferred from a single conventional evidential score.

4 Conclusion

We studied uncertainty-aware surrogate prediction for flood-transport simulation under intervention-policy shift. Across 20 training seeds, predictive accuracy and shift-detection quality were not necessarily aligned. MC Dropout achieved the strongest prediction performance, while DER showed substantially different behavior across NIG-derived uncertainty signals. In particular, ν -based uncertainty provided weak separation, whereas β -dependent quantities were considerably more informative, with $\log \beta$ giving the strongest DER shift-detection result. The evaluation is limited to one deterministic no-intervention trajectory with 77 temporally dependent samples, so the reported variation mainly reflects training randomness. Future work should evaluate these findings across multiple policies, trajectories, flood intensities, and hazard conditions.

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